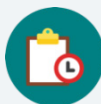
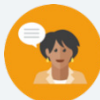


12.4 Writing Algebraic Expressions from Descriptions of Real-World Contexts

Lesson Introduction

 Benchmark	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.		 Learning Target	I can translate a real-world description into an algebraic expression.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	How can algebraic expressions represent real-world situations?			
 Lesson Narrative	This lesson builds from translating algebraic descriptions using bar models to translating real-world situations into algebraic expressions with one variable. Prior to this lesson, students will have worked with solving for unknown decimals and fractions. This lesson begins with visual representations building the student's understanding before moving away from bar models, to real-world math problems. Students practice representing real-world scenarios with algebraic expressions without the support of bar models.			
 Component Summary	12.4.1 Warm-Up (~5 min.)	Students record what they notice and wonder about an abstract image (<i>SE p. 299</i>)	12.4.4 Collaboration (10 min.)	Work collaboratively to determine whether expressions match the real-world scenarios (<i>SE p. 302</i>)
	12.4.2 Collaboration (10 min.)	Match written descriptions to bar models and expressions with a partner (<i>SE p. 300</i>)	12.4.5 Your Turn! (10 min.)	Students write algebraic expressions to represent situations (<i>SE p. 302</i>)
	12.4.3 Guided Instruction (10 min.)	Write algebraic expressions based on real-world scenarios (<i>SE p. 301</i>)	12.4.6 Wrap-Up (~5 min.)	Write a note about a mistake made during the lesson (<i>SE p. 303</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Algebraic expression		(Optional) Mini whiteboards	N/A

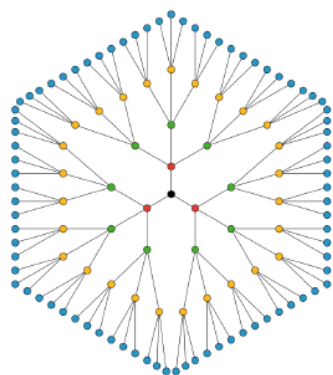
 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may try to focus on keywords as a means to figure out an algebraic expression to represent a situation. - Students may struggle to differentiate between the parts and the whole in bar models. - Students may confuse creating an algebraic expression for a situation with finding a numerical solution for a question.	- Encourage students to express the relationships in the questions in their own words and draw diagrams comparing the given quantities. Doing so will help students reason about the relationships between quantities in a situation rather than focus on the words in a question as hints to the operations needed in the expressions. - If time is an issue, then consider assigning 12.4.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	MLR 6: Three Reads Use the first read to help students understand context. Ask "What is this situation about?" After the second read, ask students "What are the quantities in the situation?" After the third read, ask students to brainstorm possible strategies to connect the situation with the appropriate expression.	MA.K12.MTR.4.1: Mathematical Discussions While pairs are working, circulate, collect, and make a visual display of vocabulary, phrases, and representations students use as they solve each situation. Make connections between how similar ideas might be communicated and represented in different ways.
 Differentiate Instruction	Provide mini whiteboards to aid students with working memory challenges. Read all situations aloud. Students who both listen to and read the information will benefit from extra processing time.	
The 12.4.6 Wrap-Up is an open-ended response in which students write a note about a mistake they made during the lesson. Consider using prompts as a way for students to assess their learning: "Describe some ways to understand how a situation can be represented mathematically. What have you learned about expressions that surprised you? Share your thoughts about using diagrams to help understand relationships. Where have you seen diagrams used previously? Where were they most helpful to you? Least helpful?"		
Lesson Notes:		

12.4.1 Warm-Up: Notice and Wonder

Component Overview: Students will be practicing their observation skills by noticing details about an abstract image. Students will also pose questions that they may be pondering about the image.



12.4.1 Warm-Up: Notice and Wonder



1. What do you notice about the image?

Answers will vary. There are dots of different colors. Lines connect all the dots. There are more dots toward the outside of the figure.

2. What do you wonder about the image?

Answers will vary. What does the image represent? Why are there different colors?



Ask students who may be interested in a challenge to write an expression that will calculate how many dots are in the image based on the number of layers shown.

12.4.2 Collaboration: Matching Activity

Component Overview: Students will collaborate to match bar models to written descriptions in addition to composing algebraic expressions to represent specific situations. Both activities review skills developed in the previous lesson.

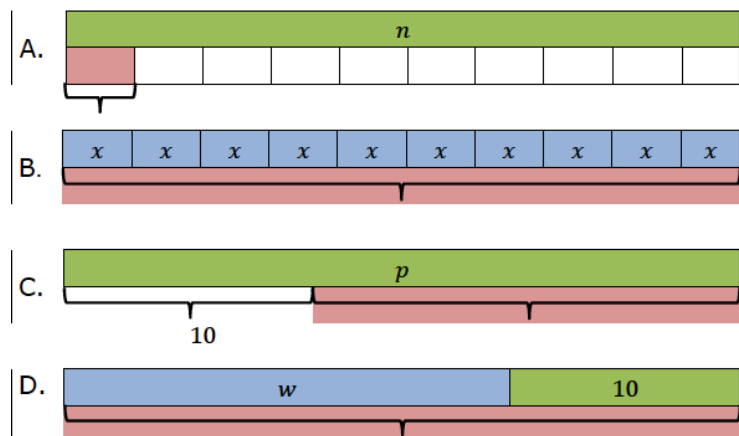


12.4.2 Collaboration: Matching Activity

Part 1

Match each written description to the appropriate bar model, where the red portion is the result of the written description. Write the letter of the bar model beside the corresponding word description.

Written Description	Bar Model
1. Ten subtracted from a number	<i>C</i>
2. A number divided by ten	<i>A</i>
3. The sum of a number and ten	<i>D</i>
4. A number times ten	<i>B</i>



Circulate as students work on 12.4.2 Collaboration, noting any gaps in learning or misconceptions that students have. Alternately, use information and observations from Lessons 1-3 to determine if small group intervention should be used during 12.4.2 Collaboration.



This section provides visual entry points through the use of bar models to represent each situation. Encourage students to think about and orally describe how A and B relate to each other to provide auditory entry points.



Activate prior knowledge to help students make connections between the written descriptions and bar models. Remind students that in a bar model, one bar represents the whole and the other bar represents the parts. Ask probing questions like:

- In C, what represents the whole? What are the parts?
- How do you know that A shows division?
- What algebraic expression could represent the bar model in B?
- What subtraction expression could D represent?

12.4.2 Collaboration: Matching Activity (continued)

Part 2

Turn and talk with your partner to compare your matches. Then, work with your partner to write an algebraic expression that represents the written description and bar model.

Written Description	Algebraic Expression
1. Ten subtracted from a number	$p - 10$
2. A number divided by ten	$\frac{n}{10}$
3. The sum of a number and ten	$w + 10$
4. A number times ten	$10x$



Students may struggle with the abstraction of variables. Consider giving students a relatable example to help them determine where the variable should go. Encourage students to substitute an actual number for the words “a number,” use that number to write the expression, and then replace the number with a variable.



Discuss how each written description is represented by an algebraic expression and a corresponding bar model. Encourage students to verbally explain the relationships seen within each grouping. For example, “10 subtracted from a number can be shown as the expression $p - 10$ because p represents the unknown number that 10 is subtracted from. In the bar model, the green bar represents the total that 10 is subtracted from. The pink bar shows the amount remaining.”

12.4.3 Guided Instruction: Writing Algebraic Expressions From Descriptions of Real-World Contexts

Component Overview: Students will write algebraic expressions to represent a given situation without using a concrete bar model. This encourages students to transition to thinking abstractly about algebraic expressions.



12.4.3 Guided Instruction: Writing Algebraic Expressions from Descriptions of Real-World Contexts

Algebraic expressions can be used to represent real-world situations.

1. Misty has 34 pairs of shoes arranged on a number of shelves. The same number of shoes are on each shelf. She has h number of shelves.
 - a. Write an algebraic expression that represents this situation.

$$\frac{34}{h}$$
 - b. What does the algebraic expression represent?
The expression represents the number of pairs of shoes on each shelf.
2. Martin buys 8 pounds of apples at a price per pound to make pie. Each pound costs p dollars.
 - a. Write an algebraic expression to represent this situation.
 $8p$
 - b. What does the algebraic expression represent?
The expression represents the total cost of the apples.
3. Paige is 6 inches taller than her brother. Paige's brother is m inches tall.
 - a. Write an algebraic expression to represent this situation.
 $m + 6$
 - b. What does the algebraic expression represent?
The expression represents Paige's height.



Three Reads

Implement a Three Reads routine here for each real-world situation.

Use the first read to help students understand context. Ask "What is this situation about?"

After the second read, ask students "What are the quantities in the situation?"

After the third read, ask students to brainstorm possible strategies to connect the situation with the appropriate expression.

Encourage students to repeat this routine themselves for each situation. This helps students connect the language in the question with the expression while keeping the intended level of cognitive demand.



If students are struggling to create algebraic expressions, allow them to use the middle step of first drawing a bar model to represent each situation. Also encourage students to verbally describe the action in each scenario before writing an algebraic expression for it. These modifications provide visual and auditory entry points.

12.4.4 Collaboration: Error Analysis

Component Overview: Students will examine and determine if algebraic expressions accurately represent a written situation. If the expressions are incorrect, students will write the correct expression and explain why the original expression is incorrect. This enables students to analytically examine algebraic expressions and learn from others' mistakes.



12.4.4 Collaboration: Error Analysis

Nero was working on a matching activity where he was given written descriptions for a series of mathematical expressions. Nero had to create an algebraic expression for each description.

- Below are three written descriptions with the algebraic expressions Nero wrote. Determine whether each expression is correct. If the expression is incorrect, explain why it isn't correct and write the correct expression.

Written Description	Nero's Expression	Correct / Incorrect Explanation
André purchased the latest gaming bundle with x new add-ons, which is 7 times as many as his cousin Jaydah has.	$x + 7$	<i>Incorrect</i> $7x$; André purchased 7 times as many, not 7 more than.
Trenton volunteers with the Future Leaders of our Youth. He was helping to collect coats for 5 families that were in need this winter. Trenton collected x coats and wanted to break up the total number of coats equally among the families.	$x - 5$	<i>Incorrect</i> $\frac{x}{5}$; Trenton is dividing x number of coats equally between 5 families.
Elena created 2 more videos for her YouTube Channel. She originally had x videos posted.	$x + 2$	<i>Correct</i>



Remind students that “error analysis” means “to look for and explain possible mistakes.” Tell students that it is not enough to just notice a mistake. Guide students to analyze why a mistake occurred in order to avoid repeating the same error in the future.



Encourage students to think about what strategies they can use to complete the error analysis. Promote metacognitive thinking by asking questions such as:

- Do you agree or disagree with that assessment? Why?
- Does anyone have the same strategy but would explain it differently?

12.4.5 Your Turn!

Component Overview: Students will practice what they have learned about creating algebraic expressions to represent specific situations. Remind students to look for the action instead of depending on keywords when formulating an expression.



12.4.5 Your Turn!

Write an algebraic expression to represent each situation.

1. Joe is 9 years younger than Drew, who is n years old.
 $n - 9$

2. Jenna's salary, s , is lowered by \$120.
 $s - 120$

3. Maria has 12 cupcakes and wants to split them among f friends.
 $\frac{12}{f}$

4. Target has 8 cookie packages with c cookies in each.
 $8c$



Consider allowing students to work in pairs and then compare their expressions with those created by another student pair. Encourage students to verbally explain why they created each algebraic expression as they did and to describe which part of an expression matches which part of the written situation.

12.4.6 Wrap-Up: Growth Mindset

Component Overview: This Wrap-Up provides students with an opportunity to reflect on their learning. Encourage students to think introspectively about the lesson, the successes they experienced, and the areas in which they felt challenged.



12.4.6 Wrap-Up: Growth Mindset

1. Write a note to yourself about a mistake you learned from today.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

*Answers will vary.
Students' notes should
address a specific mistake
related to creating an
expression for a situation.*



Consider using the following prompts as a way for students to assess their learning:

- Describe some ways to understand how a situation can be represented mathematically.
- What have you learned about expressions that surprised you?
- Share your thoughts about using diagrams to help understand relationships.
- Where have you seen diagrams used previously? Where were they most helpful to you? Least helpful?